

Multiple Linear Regression

Part II

1

MLR ... in its full potential



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2

Last class: MLR - Part I

- 1. Categorical input variables with 2 or more levels
- 2. Additive MLR: mix of categorical and continuous predictors

For example:

$$Y_i = \beta_0 + \beta_1 \text{stateWashington}_i + \beta_2 \text{povertyPercent}_i + \varepsilon_i$$

3

Last class example

$$Y_i = \beta_0 + \beta_1 \text{stateWashington}_i + \beta_2 \text{povertyPercent}_i + \varepsilon_i,$$

- For counties in Indiana,

$$Y_i = \beta_0 + \beta_1 \times 0 + \beta_2 \times \text{povertyPercent}_i + \varepsilon_i$$

$$Y_i = \beta_0 + \beta_2 \times \text{povertyPercent}_i + \varepsilon_i$$

NOTE: this is a SLR!

4

Last class example (cont.)

- For counties in Washington,

$$Y_i = \beta_0 + \beta_1 \times 1 + \beta_2 \times \text{povertyPercent}_i + \varepsilon_i$$

$$Y_i = (\beta_0 + \beta_1) + \beta_2 \times \text{povertyPercent}_i + \varepsilon_i$$

NOTE: this is another SLR! Both have the same slope (assumption of additive models)

5

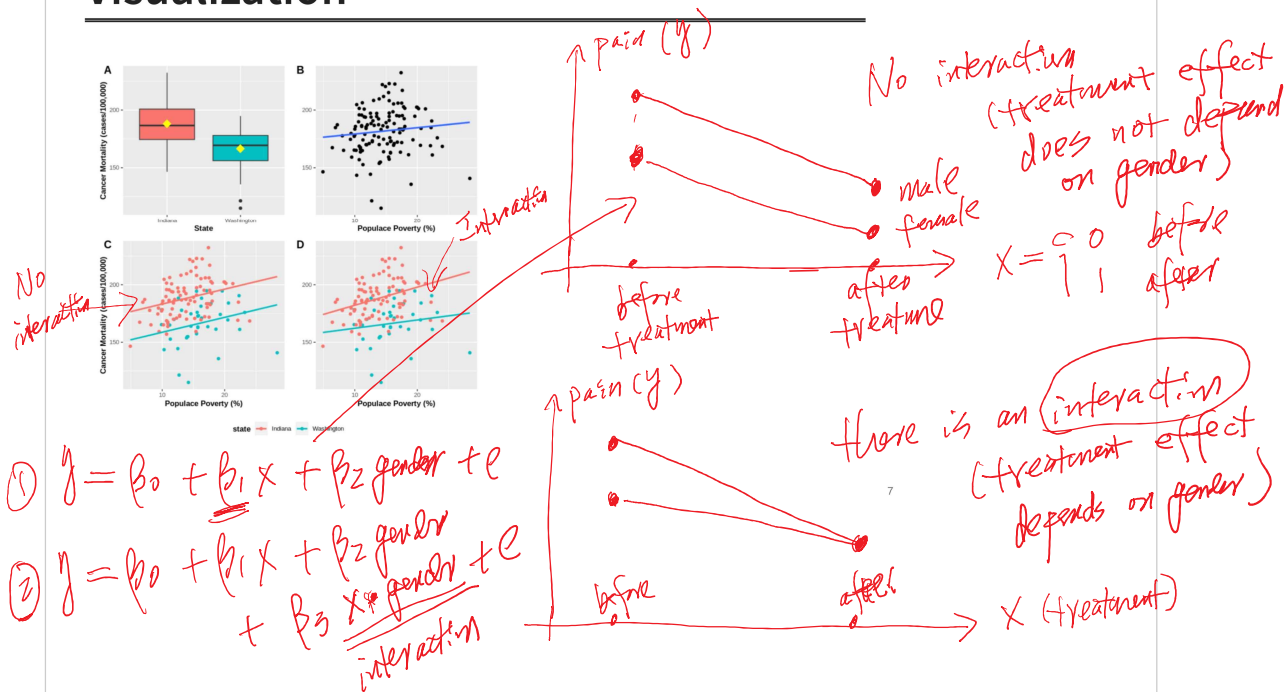
Today: MLR - Part II

- 3. MLR with interaction terms: interactions between continuous and categorical input variables
- 4. Additive MLR with any type of variables

Let's continue using the [US_cancer_data](#) to introduce other concepts of MLR.

6

Visualization



Interpretations: class discussion

- A. The average cancer mortality rate in IN is larger than that in WA
- B. We expect a positive association between cancer mortality and populace poverty percentage (on average for both states)
- C. A unit change in populace poverty percentage is **equally** associated with an increase in cancer mortality in both states. However, for any poverty percentage we expect the mortality in Indiana to be higher than that in WA
- D. A unit change in populace poverty percentage is positively associated with an increase in cancer mortality in both states but the expected change is larger in IN



3. MLR: with interaction

How can we model the relation between poverty and mortality if we believe that the change in mortality per percentage change in poverty varies by states??

If the slope changes with the levels of the categorical variable, we need to add interaction term(s)

$$Y_i = \beta_0 + \beta_1 \text{stateWashington}_i + \beta_2 \text{povertyPercent}_i + \beta_3 \text{stateWashington}_i \times \text{povertyPercent}_i + \varepsilon_i$$

interaction

$$Z = \frac{\hat{\beta}_3}{SE(\hat{\beta}_3)} \sim N(0,1)$$

If H_0 is rejected at 5% level, interaction may exist, ^{power}

Q: How do we know there is an interaction?

A: Test
 $H_0: \beta_3 = 0$ (no interaction)
 $H_1: \beta_3 \neq 0$ (there is interaction)

SLR for IN (reference)

Note that we can also think of this MLR as 2 in 1!

- For counties in Indiana, `stateWashington = 0`

$$Y_i = \beta_0 + \beta_1 \times 0 + \beta_2 \times \text{povertyPercent}_i + \beta_3 \times 0 \times \text{povertyPercent}_i + \varepsilon_i$$

$$Y_i = \beta_0 + \beta_2 \times \text{povertyPercent}_i + \varepsilon_i$$

This is a **regression line**!! It's Indiana's regression line!

10

SLR for WA

- For counties in Washington, `stateWashington = 1`

$$Y_i = \beta_0 + \beta_1 \times 1 + \beta_2 \times \text{povertyPercent}_i + \beta_3 \times 1 \times \text{povertyPercent}_i + \varepsilon_i$$

$$Y_i = (\beta_0 + \beta_1) + (\beta_2 + \beta_3) \times \text{povertyPercent}_i + \varepsilon_i$$

This is **Washington's regression line!**

Note that the intercept equals $\beta_0 + \beta_1$ and the slope equals $\beta_2 + \beta_3$

11

Interpretation:

- β_0 is intercept of the *reference* line.
- β_1 (coefficient of the dummy variable) is the difference between intercepts of both lines.
- β_2 is the slope of the reference line.
- β_3 is the the *difference* between slopes of both lines (aka *interaction* term)

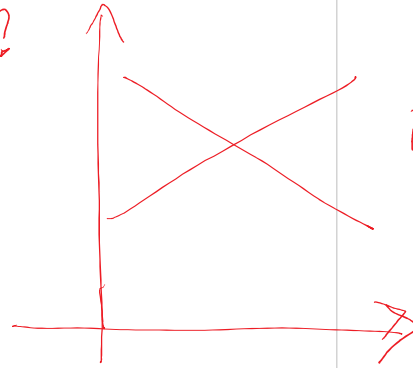
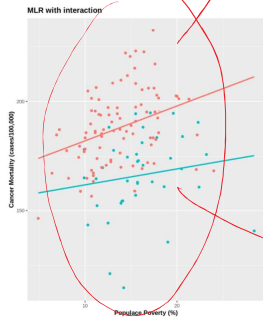
Important: using data you **estimate** these quantities!!

12

Visualization of MLR with interaction

MLR with interaction

VISUALIZATION OF MLR WITH INTERACTION



interaction strong

NOTE: the model with interaction has 4 coefficients to characterize 2 lines. We do NOT assume a common slope!!

SRS

maybe large

$$z = \frac{\beta_3}{SE(\beta_3)}$$

⇒ z value may be small
⇒ p-value may be large
⇒ H_0 may not be rejected

Code

```
MLR_stat_poverty_int <- lm(TARGET_deathRate ~ state * povertyPercent, data = WI_cancer_data)
```

variables are added to the model using *

```
MLR_stat_poverty_int <- tidy(lm(TARGET_deathRate ~ state * povertyPercent, data = WI_cancer_data)) %>% mutate_if(is.numeric, round, 3)
```

MLR_stat_poverty_int

A tibble: 4 x 5

term	estimate	std.error	statistic	p.value
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
(Intercept)	166.379	6.768	24.584	0.000
stateWashington	-11.451	13.176	-0.869	0.386
povertyPercent	1.577	0.474	3.327	0.001
stateWashington:povertyPercent	-0.855	0.833	-1.027	0.306

Plot D illustrates this case

interaction β_3
 $SE(\hat{\beta}_3)$
 $\beta_3 / SE(\hat{\beta}_3)$

p-value = 0.306 > 0.05,
so there is no strong evidence for interaction (i.e. weak interaction).
Thus, we can remove the interaction term from the model

Interpretation: The relationship between poverty and mortality does not depend on state.

Hypothesis Testing

As before, we can use the estimated parameters to make inference about the population parameters.

The null hypotheses are as usual:

$$H_0 : \beta = 0$$

$$H_1 : \beta \neq 0$$

Note: For simplicity, I've omitted the subindex of β but we can set a test for each coefficient in the model (or contrasts).

In R

The `tidy` table reports results to test these hypotheses for each parameter in the regression!!

Example

```
MLR_state_poverty_int <- tidy(lm(TARGET_deathRate ~ state * povertyPercent,  
data = WI_cancer_data)) %>% mutate_if(is.numeric, round, 3)
```

MLR_state_poverty_int

A tibble: 4 × 5

term	estimate	std.error	statistic	p.value
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
(Intercept)	166.379	6.768	24.584	0.000
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16

The last row tests the following hypothesis:

$$H_0 : \beta_3 = 0 \quad \text{vs} \quad H_1 : \beta_3 \neq 0$$

where β_3 represents the slope of WA minus the slope of IN (difference of slopes)

Since the p -value is larger than 0.05, there is not enough statistical evidence to reject the null hypothesis that the association between change in mortality per unit percentage change in poverty is the same in both states (same slopes).

17

MLR ... in its full potential

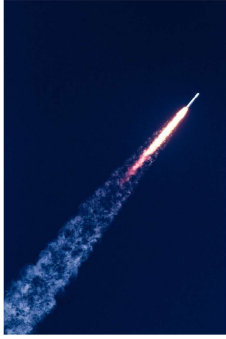


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18

Modelling many variables with MLR

← multiple linear regression

- In a MLR the response is always a continuous variable
- The predictors can be either discrete or continuous
- We can include interaction terms for some or all variables
- We may prefer to use additive models, even if their assumptions may not be realistic

19

In practice ...

Additive models are more common in practice since they are easier to interpret, in particular when many variables are available.

Assumption of additive models

The association between the change of the response per unit change in another variable is the same, regardless of the values of other variables in the model.

20

Examples of additive assumption:

- the increase in calories burned per an additional hour of exercise is the same for athletes of different ages.
- the increase in sale price per additional square foot of a house is the same, regardless of the location.
- the increase in mortality per additional percentage of populace poverty does not change with the percentage of people with private coverage.

Even when this assumption may not be realistic, you may prefer to simplify the model.

21

Interpretation for additive models

In additive models, you interpret each coefficient separately while “holding all other variables constant”.

since the model is additive, it doesn't matter at which value the variables are held constant!

For example:

- A unit increase in populace poverty is associated with an expected increase of 1.3 in mortality in both WA and IN. But for any populace poverty percentage, the cancer mortality rate in IN is higher than that in WA

22

Interpretation of models with interaction

In models including interaction terms, you can't interpret each coefficient separately while “holding all other variables constant”

This is a common mistake!

The association between variables changes with the values of other variables in the model

We will test you only on the types of models with interaction covered in lectures

23

What have we learned today?

- We can add any type of input variable in a LR
- The interpretation of the coefficients depends on the type of variable and model

Some particular cases:

- Additive LR with one continuous and one categorical variable: multiple parallel lines
- Interaction LR with one continuous and one categorical variable: multiple non-parallel lines

24

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- Additive LR with multiple continuous models: hyperplanes

In general, for **additive models**:

- you interpret each coefficient maintaining all other variables in the model constant (at any value) *or adjusted for other variable*

Hypothesis tests can be used to test if each parameter in the regression equals 0 (null hypothesis)

- `lm` computes SE and p-values using results from classical theory

25